

Isomorphism Theorems (동형정리)

$\phi: G \rightarrow G'$ 을 group homomorphism

이라 하면

$G/\text{Ker } \phi \cong \phi[G]$ 가 성립.

$$g \text{Ker } \phi \mapsto \phi(g)$$

Theorem 1.

$N \trianglelefteq G$, $\gamma: G \rightarrow G/N$ ($g \mapsto gN$)

$\left\{ \begin{array}{l} N \text{을 포함하는} \\ G \text{의 정규부분군} \end{array} \right\} \xleftrightarrow[\text{일대일 대응 존재.}]{\quad} \left\{ \begin{array}{l} G/N \text{의} \\ \text{정규부분군} \end{array} \right\}$

$N \leq L \trianglelefteq G$ $\phi: L \rightarrow \gamma[L] = \phi(L)$

증명.

If $N \leq L$, $\gamma^{-1}[\gamma[L]] = L$.

먼저 $L \subseteq \gamma^{-1}[\gamma[L]]$ 은 clear 하다.

$x \in \gamma^{-1}[\gamma[L]]$ 이라 하자. $\gamma(x) \in \gamma[L]$

$\exists l \in L, \gamma(x) = \gamma(l) \Rightarrow \gamma(xl^{-1}) = 1.$

$\Rightarrow xl^{-1} \in L \Rightarrow x \in lL = L.$

If $\phi(L_1) = \phi(L_2)$, then $\gamma[L_1] = \gamma[L_2].$

$(N \leq L_1, N \leq L_2). \Rightarrow \gamma^{-1}[\gamma[L_1]] = \gamma^{-1}[\gamma[L_2]]$

$\Rightarrow L_1 = L_2.$

$\Rightarrow \phi : \text{Injective}.$

Claim. $H \trianglelefteq G/N, \gamma[\gamma^{-1}[H]] = H$

($\Leftarrow, \phi[\gamma^{-1}[H]] = H \Rightarrow \phi : \text{onto}$)

$\gamma[\gamma^{-1}[H]] \subseteq \text{clear. } H \subseteq \gamma[\gamma^{-1}[H]].$

Theorem 2.

Every normal subgroup of G/N can be uniquely expressed as L/N for some G .

Every subgroup of G/N can be uniquely expressed as L/N for some $N \leq L \leq G$.

$N \trianglelefteq \bar{G}$, $\bar{A} = A/N$ for $N \leq A \leq G$.

① $A \leq B \iff \bar{A} \leq \bar{B}$

② If $A \leq B$, $(B:A)(A:B)$

If $H, N \leq G$, let $HN = \{hn \mid h \in H, n \in N\}$
 HN is not necessarily subgroup of G .

ex) $G = S_3$.

$H = \{e, (1, 2)\}$, $N = \{e, (1, 3)\}$.

$HN = \{e, (1, 3), (1, 2), (1, 3, 2)\} \neq G_3$

Theorem 3.

$H, N \leq G$. H 와 N 이 G 의 정규부분군이다.

$\Rightarrow HN \leq G$.

증명.

$H \trianglelefteq G$. ① HN of $e \cdot e = e \in HN$. $HN \neq \emptyset$

② $h_1 n_1, h_2 n_2 \in HN$ ($h_1, h_2 \in H, n_1, n_2 \in N$)

$$h_1 n_1 (h_2 n_2)^{-1} = h_1 n_1 n_2^{-1} h_2^{-1}$$

$$= h_1 n_1 n_2^{-1} h_2^{-1} n_1 n_2$$

$$= h_1 n_1 n_2^{-1} h_2^{-1} (n_1 n_2^{-1})^{-1} n_1 n_2^{-1} \in HN$$

Theorem 4.

$H, N \trianglelefteq G$, then $HN \trianglelefteq G$.

증명) We know that $HN \leq G$.

$$\forall g \in G, g h n g^{-1} = (g h g^{-1})(g n g^{-1}) \in HN$$

$$\therefore HN \trianglelefteq G.$$

Theorem 5. 제 2 동형정리.

$H \leq G$, $N \trianglelefteq G$. 그러면 $HN/N \cong H/H \cap N$.

증명.

$f: H \rightarrow HN/N$ 이라 하자. $h \mapsto hN$.

$$f(h_1 h_2) = h_1 h_2 N = (h_1 N)(h_2 N) = f(h_1) f(h_2).$$

f 는 전사인 준동형 사상이 된다.

$$\text{Ker } f = \{h \in H \mid f(h) = hN = N\} \Rightarrow h \in N.$$

$$\therefore \text{Ker } f = H \cap N.$$

따라서 $H/H \cap N \cong HN/N$. $\rightarrow$ 제 1 동형정리

Theorem 5. 제 3 동형정리.

$H, K \trianglelefteq G$. $K \leq H$. Then

$$(G/K)/(H/K) \cong G/H.$$

증명.

$\xi : G/K \rightarrow G/H$ 와 하자.

$$gK \mapsto gH$$

If $g_1K = g_2K$, $g_1^{-1}g_2 \in K \leq H$.

so $g_1H = g_2H \Rightarrow \xi$ is well-defined.

$$\begin{aligned}\xi(g_1K \cdot g_2K) &= \xi(g_1g_2K) = g_1g_2H \\ &= (g_1H)(g_2H) \\ &= \xi(g_1K) \xi(g_2K)\end{aligned}$$

$\rightarrow \xi : \text{Homomorphism.}, \xi$ is surjective.

$$\begin{aligned}\text{Ker } \xi &= \{gK \in G/K \mid gH = H\} \\ &= \{gK \in G/K \mid g \in H\} = H/K\end{aligned}$$

Definition. Subnormal Series.

$H_0, H_1, H_2, \dots, H_n$ 이

$\{e\} = H_0 \triangleleft H_1 \triangleleft H_2 \dots \triangleleft H_n = G$ 가 되면

$\{H_i\}_{i=1}^n$ 을 G 의 subnormal series 라고 한다.
만일 $H_i \trianglelefteq G$ 라면 $\{H_i\}$ 를 normal series

Remark. Normal series 는 Subnormal series
지만 역은 성립하지 않는다.

ex) $D_4 = \{e, r, r^2, r^3, s, rs, r^2s, r^3s\}$

$\{e\} \triangleleft \{e, s\} \triangleleft \{e, r^2, s, r^2s\} \triangleleft D_4$
↳ Not normal

ex) $\{0\} \triangleleft 6\mathbb{Z} \triangleleft 3\mathbb{Z} \triangleleft \mathbb{Z}$.

$\{0\} \triangleleft 8\mathbb{Z} \triangleleft \mathbb{Z}$.

Definition. Refinement $\rightarrow$ 더 잘게 쪼갠다는 느낌.

A subnormal series $K_0 \triangleleft K_1 \triangleleft \dots \triangleleft K_m = G$

is a refinement of a subnormal series

$H_0 \triangleleft H_1 \triangleleft H_2 \dots \triangleleft H_n = G$ if for every i , $H_i = K_j$

for some j .

$\exists j$. $\{H_0, H_1, \dots, H_n\} \subseteq \{K_0, K_1, \dots, K_m\}$

ex). $\{0\} \triangleleft 12\mathbb{Z} \triangleleft 6\mathbb{Z} \triangleleft 3\mathbb{Z} \triangleleft \mathbb{Z}$
 는 $\{0\} \triangleleft 12\mathbb{Z} \triangleleft 3\mathbb{Z} \triangleleft \mathbb{Z}$ 의 refinement이다.

Definition.

두 개의 subnormal series $\{H_i\}$, $\{K_j\}$ 에 대해
 isomorphic 하다고 한다.

$\rightarrow \{H_{i+1}/H_i : i\} \longleftrightarrow \{K_{j+1}/K_j : j\}$ 사이에 일대일
 대응이 있고, 각각의 대응으로 연결되는 순서끼리 동형이다.

ex), $\{0\} \triangleleft \{0, 3\} \triangleleft \mathbb{Z}_6$
 $\{0\} \triangleleft \{0, 2, 4\} \triangleleft \mathbb{Z}_6$.

$\mathbb{Z}_6 / \{0, 3\} \cong \mathbb{Z}_3$, $\{0, 3\} / \{0\} \cong \mathbb{Z}_2$.

$\{0, 2, 4\} / \{0\} \cong \mathbb{Z}_3$, $\mathbb{Z}_6 / \{0, 2, 4\} \cong \mathbb{Z}_3$.

Theorem 6.

Two subnormal series of a common group G have
 isomorphic refinements.

ex) $\{0\} \triangle \mathbb{Z} \triangle 4\mathbb{Z} \triangle \mathbb{Z}$
 $\{0\} \triangle 6\mathbb{Z} \triangle 3\mathbb{Z} \triangle \mathbb{Z}$ } Not Isomorphic.
 $\mathbb{Z} \quad \mathbb{Z}_2 \quad \mathbb{Z}_3 \quad \mathbb{Z}_2 \quad \mathbb{Z}_4$
 $\{0\} \triangle 48\mathbb{Z} \triangle 24\mathbb{Z} \triangle 8\mathbb{Z} \triangle 4\mathbb{Z} \triangle \mathbb{Z}$
 $\Rightarrow \{0\} \triangle 48\mathbb{Z} \triangle 24\mathbb{Z} \triangle 6\mathbb{Z} \triangle 3\mathbb{Z} \triangle \mathbb{Z}$
 $\mathbb{Z} \quad \mathbb{Z}_2 \quad \mathbb{Z}_4 \quad \mathbb{Z}_2 \quad \mathbb{Z}_3$

$$\{e\} = H_0 \triangle H_1 \triangle H_2 \triangle \dots \triangle H_n = G.$$

$$\{e\} = K_0 \triangle K_1 \triangle K_2 \triangle \dots \triangle K_m = K.$$

$$H_{i,0} = H_i(H_{i+1} \cap K_0) \triangle H_i(H_{i+1} \cap K_1) \triangle H_i(H_{i+1} \cap K_2) \\ \dots \triangle H_i(H_{i+1} \cap K_m) = H_{i+1} = H_{i,m}$$

$$H_0 = H_{0,0} \triangle H_{0,1} \triangle \dots \triangle H_{0,m-1} \\ \triangle H_{1,0} \triangle H_{1,1} \triangle \dots \triangle H_{1,m-1} \\ \vdots \\ \triangle H_{n-1,0} \triangle H_{n-1,1} \triangle \dots \triangle H_{n-1,m-1} \triangle H_n.$$

$$H_{i,j+1}/H_{i,j} \cong K_{j,i+1}/K_{j,i}$$

Definition. Composition Series.

$\{H_i\}$ 가 subnormal series 라 하자.

모든 i 에 대해 H_{i+1}/H_i 가 simple 이면,

$\{H_i\}$ 를 composition series 라고 한다.

$\{e\} \triangle A_n \triangle S_n$: Composition Series of S_n .